Bounded Describability and Structural Uniqueness: The Axiom of Uniqueness as a Universal Theorem in Dynamical Systems

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Abstract

This paper formalizes the transition of the Axiom of Uniqueness—originally conceived as an operational postulate within the Invariant Integrity Audit Engine (MAII)—into a fundamental theorem at the intersection of metric topology and algorithmic information theory. We demonstrate that attractor uniqueness is not an arbitrary constraint imposed upon a system, but an inescapable logical and mathematical consequence of the condition of describability. By formalizing the Space of Bounded Describability (Ω_D), we prove that any dynamics whose trajectory can be represented by a finite amount of information necessarily possesses a non-empty, compact, and invariant ω-limit set, uniquely determined by the trajectory. Consequently, the Axiom of Uniqueness is formally derived as a universal theorem: every describable dynamics structurally converges, and uncontained divergence is mathematically excluded from the space of representation.

1 Introduction: From Postulate to Theorem

In the development of containment architectures for stochastic systems and generative models, the MAII/IDICOC framework has operated under the so-called "Axiom of Uniqueness", assuming that every valid trajectory collapses into an invariant core. However, postulating uniqueness as an internal axiom obscures its true topological nature.

The central thesis of this paper is that structural uniqueness does not need to be postulated; it is a demonstrable property that emerges from a more primitive constraint: describability. If a dynamics is imaginable, encodable, or computable by a finite information system over time, it is subject to strict limits of descriptive entropy.

The scope of Ω_D is not limited to engineered systems. By the Church-Turing thesis, any process that can be imagined, described in a finite symbolic language, or represented by any cognitive or formal system admits a Turing-computable encoding. Consequently, the condition of bounded Kolmogorov complexity—the defining criterion of Ω_D—is satisfied by every dynamic that exists operatively within any representable universe. A dynamic that violates this condition is not merely divergent; it is, by definition, outside the boundary of the conceivable.

2 Formalization of the Space of Describability (Ω_D)

To rigorously link the geometry of the phase space with computability theory, we define the dynamics over a metric space and establish an encoding that preserves the distinguishability of states.

Definition 1 (Dynamical System and ϵ-faithful Encoding). *Let (S, d) be a complete metric space. We define a discrete-time dynamical system via the continuous, TTE-computable map $\xi : S \to S$. The trajectory of an initial state $s \in S$ is $\mathcal{O}^+(s) = \{\xi^t(s)\}_{t=0}^{\infty}$.*

For any resolution threshold $\epsilon > 0$, we define an ϵ-faithful algorithmic description function $\delta_\epsilon : S \to \{0,1\}^*$. We assume δ_ϵ is an effective encoding and is monotonic: for $\epsilon_1 < \epsilon_2$, $\delta_{\epsilon_1}(s)$ is a computable refinement of $\delta_{\epsilon_2}(s)$, satisfying $K(\delta_{\epsilon_2}(s)) \leq K(\delta_{\epsilon_1}(s)) + \mathcal{O}(1)$. Furthermore, δ_ϵ must guarantee injectivity for topologically ϵ-separated states:

$$\forall s, s' \in S : d(s, s') \geq \epsilon \implies \delta_\epsilon(s) \neq \delta_\epsilon(s')$$

$K(\cdot)$ denotes the prefix Kolmogorov complexity.

Definition 2 (The Space of Bounded Describability Ω_D). *Ω_D is the subset of initial states whose trajectories require information bounded uniformly over time:*

$$\Omega_D = \left\{ s \in S \mid \forall \epsilon > 0, \exists C_\epsilon \in \mathbb{N}, \forall t \geq 0,\ K(\delta_\epsilon(\xi^t(s))) \leq C_\epsilon \right\}$$

Since ξ is a TTE-computable map, $K(\delta_\epsilon(\xi(s))) \leq f(K(\delta_\epsilon(s))) + \mathcal{O}(1)$ for a computable f, guaranteeing that updates do not spontaneously generate infinite structural entropy.

3 Central Lemma: Bounded Complexity Implies Precompactness

Lemma 1. *If an initial state s belongs to Ω_D, then its orbit $\mathcal{O}^+(s)$ is totally bounded in (S, d), and its closure $\overline{\mathcal{O}^+(s)}$ is precompact.*

Proof. Assume $\mathcal{O}^+(s)$ is not totally bounded. Then there exists $\epsilon > 0$ and an infinite sequence $\{s_{t_i}\}$ such that $d(s_{t_i}, s_{t_j}) \geq \epsilon$. By Definition 1, δ_ϵ assigns distinct binary strings to these states. Since $s \in \Omega_D$, $K(\delta_\epsilon(s_{t_i})) \leq C_\epsilon$. The number of prefix-free programs of length at most C_ϵ is bounded by $2^{C_\epsilon+1}$ [1]. This contradicts the infinite set of distinct strings. Thus, $\mathcal{O}^+(s)$ is totally bounded and $\overline{\mathcal{O}^+(s)}$ is precompact. $\square$

4 Main Theorem: The Limit of Describable Reality

Theorem 1 (Theorem of Bounded-Description Complexity). *If $s \in \Omega_D$, its ω-limit set $\omega(s) = \bigcap_{n \geq 0} \overline{\bigcup_{t \geq n} \xi^t(s)}$ is non-empty, compact, and strictly invariant under ξ.*

Proof. Since $X = \overline{\mathcal{O}^+(s)}$ is compact:

1. **Non-empty:** Every infinite sequence in X has a convergent subsequence, whose limit is in $\omega(s)$.

2. **Compact:** $\omega(s)$ is the intersection of nested closed sets in X, hence compact.

3. **Invariant:** For the inclusion $\xi(\omega(s)) \subseteq \omega(s)$: if $x \in \omega(s)$, there exists $t_k \to \infty$ with $\xi^{t_k}(s) \to x$. By continuity of ξ, $\xi^{t_k+1}(s) \to \xi(x)$, so $\xi(x) \in \omega(s)$. For the reverse inclusion $\omega(s) \subseteq \xi(\omega(s))$: let $y \in \omega(s)$. The sequence $\{\xi^{t_k-1}(s)\}$ lies in the compact set X and has a convergent subsequence $\xi^{t_{k_j}-1}(s) \to x \in \omega(s)$. By continuity, $\xi(x) = y$, so $y \in \xi(\omega(s))$.

□

Remark on Semicontinuity. If ξ is locally Lipschitz, the map $s \mapsto \omega(s)$ is upper semicontinuous with respect to the Hausdorff metric.

Observation 1 (Trajectory-wise vs. Global Uniqueness). The set $\omega(s)$ is uniquely determined by s and ξ. While this implies trajectory-wise uniqueness, it does not imply global uniqueness (a system may possess multiple disjoint attractors).

5 Corollary: The Derivation of the MAII Axiom

Corollary 1 (Derivation of Structural Uniqueness)**.** *Immediate from Theorem 1: every $s \in \Omega_D$ possesses a non-empty compact invariant ω-limit set, satisfying the structural uniqueness required by the MAII.*

The IDICOC framework does not invent uniqueness; it recognizes it as a requirement for existence. When a trajectory approaches $\partial\Omega_D$, the MAII MUX executes a Geodesic Interception, forcing the system to a pre-compiled safe attractor. This ensures global convergence.

6 Characterization of Counterexamples ($\notin \Omega_D$)

The analytical power of the Theorem is evidenced by characterizing the dynamics expelled from Ω_D. Brudno's theorem [3] establishes that in ergodic systems, complexity grows for μ-almost every point if $h_{KS} > 0$. Our framework provides the topological link:

- **Positive KS Entropy:** If $h_{KS} > 0$, almost all trajectories have unbounded descriptive complexity. While periodic orbits may stay in Ω_D, chaotic dynamics with $h_{KS} > 0$ are excluded from Ω_D because, even when spatially bounded, the informational complexity of describing their trajectories grows without bound, violating the uniform bound C_ϵ required by Definition 2.

- **Spatial Divergence:** Linear expansion requires $K \to \infty$ to maintain resolution, failing precompactness and falling outside Ω_D.

7 Remarks and Extensions: Coalgebraic Foundations

While the present paper establishes the topological and information-theoretic grounds of Ω_D, the framework naturally extends into universal coalgebra, which will be the subject of future formalization. Conceptually, we define the Coalgebraic Exterior as $\Omega_D^c = S \setminus \Omega_D$. A state $s \in \Omega_D^c$ iff its coalgebraic trace requires an unbounded injection of information. In categorical terms, Ω_D

acts as the maximal subcoalgebra admitting a unique homomorphism (!) into the final coalgebra of bounded behaviors.

In universal coalgebra, the Finality Axiom dictates that from a valid state space into a final coalgebra, there exists exactly one unique homomorphism (!). Within IDICOC, this maps to Ω_D. For $s \in \Omega_D$, $!_{\Omega_D}$ is well-defined. For $s \in \Omega_D^c$, uniqueness is mathematically undefined. Thus, the MAII framework operates by mechanically enforcing this coalgebraic boundary. Providing the rigorous categorical proofs for Ω_D as a maximal subcoalgebra remains a designated direction for future theoretical work.

8 Conclusion

This paper redefines the paradigm of systemic validity in computational environments. We have demonstrated that the Axiom of Uniqueness is not a premise, but the central theorem of Bounded Describability. If a dynamics exists and is representable by finite information, it must be geometrically precompact and converge to an invariant attractor. By founding the IDICOC framework upon this analytical proof, we endow the Invariant Integrity Audit Engine with a definitive mathematical justification: deterministic containment and the suppression of divergent noise are not mere engineering choices, but mathematical imperatives required to sustain any computationally structured reality.

References

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